ChatGPT-Assisted Learning Effectiveness and Academic Achievement: a Mechanism-Based Model in Higher Education

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Abstract

This study examines the impact of ChatGPT-assisted learning on the academic achievement of hospitality and tourism students in Egyptian public universities, with particular emphasis on the mediating roles of perceived usefulness and self-regulated learning. Drawing conceptually on the Technology Acceptance Model (TAM), the study adopts a contextualized framework that emphasizes perceived usefulness while incorporating ChatGPT-assisted learning effectiveness as a learning-oriented driver within generative AI-supported educational environments. A quantitative research design was employed using an online survey administered to students who actively used ChatGPT for academic purposes. A total of 689 valid responses were collected from nine public universities and analyzed using Partial Least Squares Structural Equation Modeling (PLS-SEM) to test the proposed hypotheses. The findings indicate that ChatGPT-Assisted Learning Effectiveness (CALE) has a statistically significant and positive direct effect on academic achievement (AA; beta = 0.386, T = 3.946, $p < 0.001$, 95% CI = 0.192–0.561) and strongly predicts perceived usefulness (beta = 0.673, T = 9.274, $p < 0.001$, 95% CI = 0.581–0.742) and self-regulated learning (beta = 0.707, T = 10.734, $p < 0.001$, 95% CI = 0.621–0.779). In turn, PU (beta = 0.281, T

= 3.854, $p < 0.001$, 95% CI = 0.142–0.417) and SRL (beta = 0.220, T = 2.418, $p = 0.016$, 95% CI = 0.041–0.356) significantly enhance academic achievement. Mediation analyses further confirm that PU (beta = 0.189, T = 2.366, $p = 0.018$, 95% CI = 0.031–0.284) and SRL (beta = 0.156, T = 3.699, $p < 0.001$, 95% CI = 0.102–0.301) partially mediate the relationship between CALE and academic achievement. These findings offer important theoretical insights by contextualizing TAM's performance-related logic within generative AI-driven learning environments and refining its application to academic outcome settings, while highlighting self-regulated learning as a critical explanatory mechanism. From a practical perspective, the study provides valuable implications for educators and policymakers by emphasizing the need to promote students' perceived usefulness of ChatGPT and foster learner autonomy, positioning generative AI as a powerful pedagogical support tool for enhancing academic success in hospitality and tourism education.